

Dataset Collection and Visualization

Introduction

In this phase, the dataset required for the Human Development Index (HDI) Prediction project is identified and collected from Kaggle. After downloading the dataset, its structure, features, and quality are examined to understand the information it contains. This initial analysis helps determine whether preprocessing is required before training the machine learning model.

Steps Performed

- Search for a suitable Human Development Index (HDI) dataset on Kaggle.
- Download the dataset in CSV format.
- Load the dataset into Python using the Pandas library.
- Examine the dataset by checking its shape, column names, data types, and summary statistics.
- Identify missing values, duplicate records, and inconsistent data.
- Visualize the dataset using charts such as histograms, bar charts, scatter plots, box plots, and correlation heatmaps to better understand patterns and relationships among variables.

Outcome

By the end of this phase, the dataset is successfully collected, understood, and visually explored. The insights gained from visualization support feature selection, data preprocessing, and the development of an accurate HDI prediction model.